CVPR
2016
BEST PAPER

Deep Residual Learning for Image Recognition

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Paper

Problem

Plain deep networks suffer a **degradation problem**: past a depth threshold, training accuracy **saturates and gets worse** — an optimization failure on very deep stacks, not overfitting from added capacity.

> *A 56-layer plain net has higher training error than a 20-layer one — depth itself starts to hurt.*

Motivation

- Depth matters, but **vanishing/exploding gradients** alone — already mitigated by BN and He init — don't explain the new degradation.
- A deeper net *could* match a shallower one by identity-padding the extras — yet plain stacks fail.
- Normalized init + BN + warm-up can't push past the **~30-layer wall**; a structural reformulation is needed.

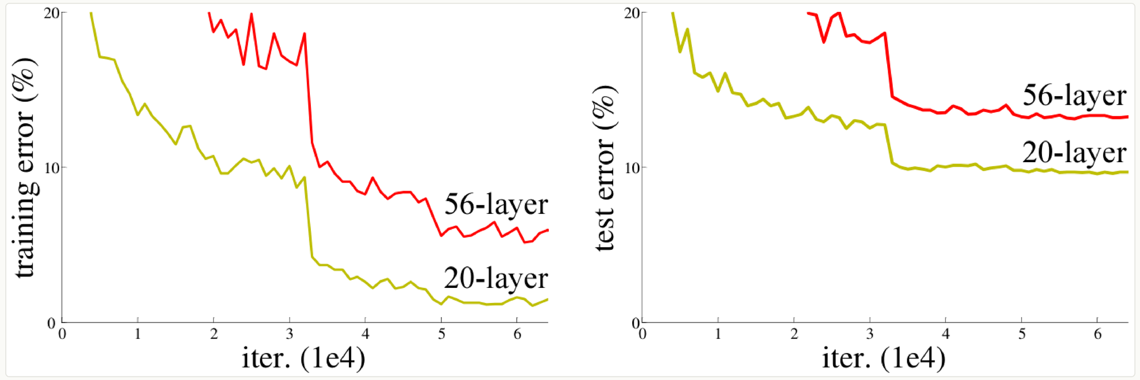

Figure 1. CIFAR-10 error: 56-layer plain net is worse than the 20-layer one — degradation, not overfitting.

Method

- Each block learns a **residual**; the input is added back through a **parameter-free identity shortcut**, adding no extra parameters and almost no compute.
- When dimensions change, choose **zero-padding** (Option A) or a **1×1 projection** (Option B).
- **Bottleneck** blocks (1×1 → 3×3 → 1×1) make 50/101/152-layer nets cheaper than VGG.
- Trained with **SGD** (batch 256, lr 0.1 with step decay), **BN** after each conv, **He init**, weight decay 1e-4 — no dropout anywhere.

$$\mathbf{y} = F(\mathbf{x}, \{W_i\}) + \mathbf{x}$$

F is a stack of 2–3 weight layers; the + is element-wise addition through a parameter-free shortcut.

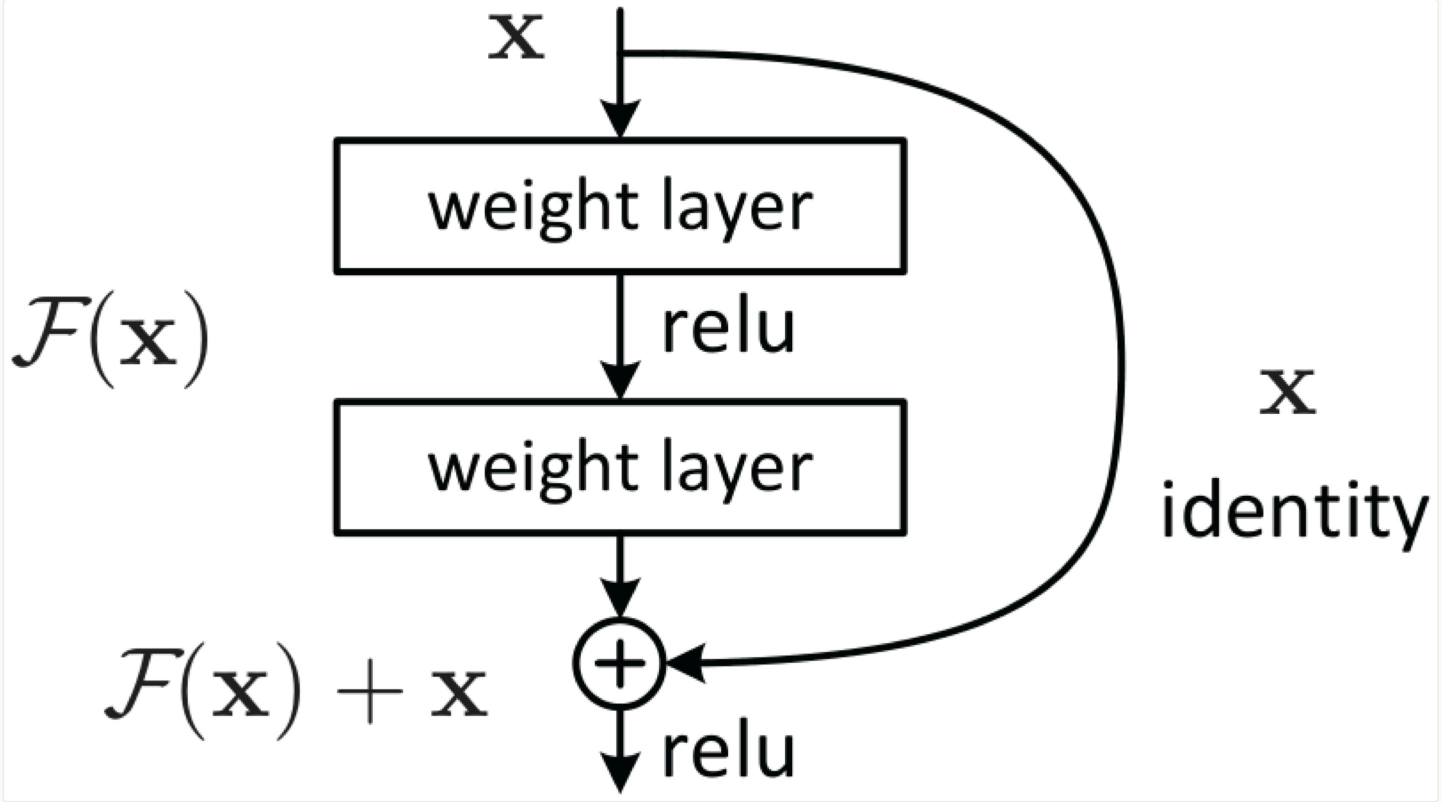

Figure 2. Residual learning: a building block — the residual F(x) is added to an identity shortcut x.

2-layer block · 3-layer bottleneck · Identity shortcut · 1×1 projection · BN · He init · SGD 256

Key Results

IMAGENET VAL (10-CROP)	TOP-1	TOP-5
Plain-34	28.54	10.02
ResNet-34	25.03	7.76
ResNet-50	22.85	6.71
ResNet-101	21.75	6.05
ResNet-152	**21.43**	**5.71**

−3.51 pts top-1: Plain-34 → ResNet-34 — degradation cleared.

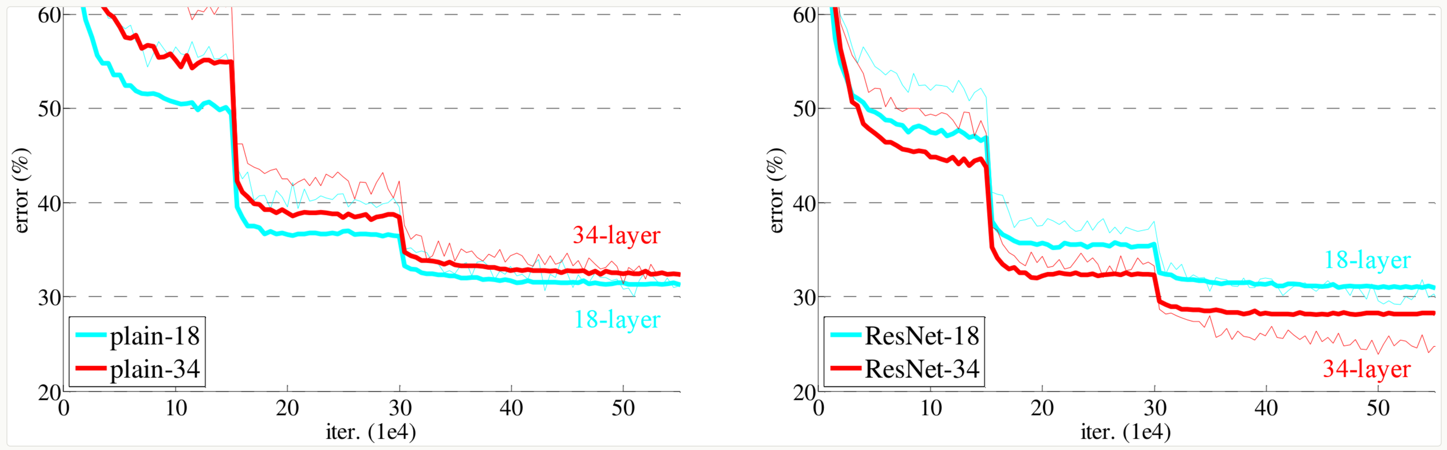

Figure 4. ImageNet training curves. Thin = train, bold = val. ***Left:*** *plain — deeper is worse.* ***Right:*** *ResNet — deeper is better.*

+6.0 COCO mAP@[.5,.95]	**+28%** relative gain	**5×** ILSVRC + COCO 2015 wins

SO WHAT → *A single 152-layer ResNet beats every prior* ***ensemble*** *on ImageNet, and the 6-model ResNet ensemble wins ILSVRC 2015 at* ***3.57%*** *top-5. The same ResNet-101 backbone, dropped into Faster R-CNN with no architectural changes, also powers the COCO and ImageNet detection / localization winners — a backbone-level win, not a task-specific trick. ResNet became the default vision backbone for years.*

Headline Numbers

3.57%
ImageNet top-5 · ILSVRC 2015 winner
152-layer ResNet ensemble — 1st place

4.49% single-model top-5	**+28%** COCO mAP gain	**152** layers (8× VGG)

Ablation Study

- **Shortcut type** (ResNet-34 top-1): A 25.03 → B 24.52 → C 24.19. Identity shortcuts are enough; projections everywhere add little.
- **1202-layer** CIFAR-10 ResNet trains to <0.1% error — no optimization difficulty at 10^3 layers.

Takeaway

Residual learning with identity shortcuts **removes the degradation barrier** — accuracy keeps improving as depth grows into the hundreds.

Depth + the right structural prior is a clear win.

The residual connection became a building block reused across deep learning — **Transformers**, diffusion, and beyond.